

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Faculty of Technology and Engineering
Devang Patel Institute of Advance Technology and Research (DEPSTAR)
Department of Information Technology
ITUC301: Data Engineering

PRACTICAL - 1

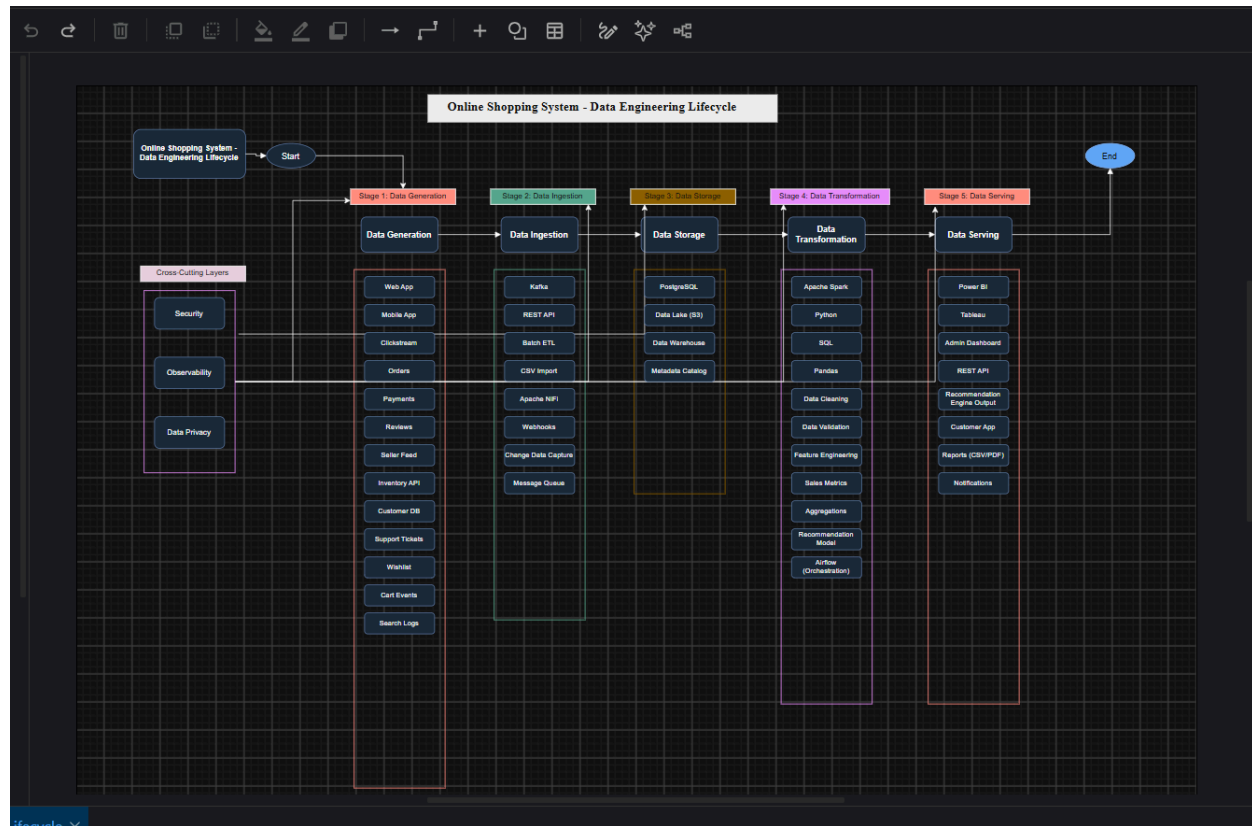
Title : Understanding the Data Engineering Lifecycle

Problem Definition: Study, dissect, and map the complete end-to-end data engineering lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: data generation, ingestion, storage, transformation, and serving.

The solution requires creating a detailed architectural data flow diagram along with a system manifesto documenting cross-cutting lifecycle undercurrents (security, observability, and data privacy). This initial structure must be committed to a personal Git repository.

Dataset/Test Data: Mock enterprise e-commerce dataset schema definition (containing transactional database entities, clickstream tracking JSON payloads, and third-party inventory log streams).

OUTPUT :



Key Questions :

1: How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?

→ Downstream serving needs (e.g. real-time recommendations, live dashboards) determine whether ingestion must be streaming vs batch, and whether storage should prioritize fast query access (warehouse) or cheap bulk storage (data lake). These serving-layer demands are considered even during generation, so the right data format and granularity is captured from the start.

2: Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

→ The broader data lifecycle covers data's entire journey across an organization — creation, business use, governance, archival, and deletion. The technical data engineering lifecycle is the narrower subset engineers

directly build and operate: ingestion, storage, transformation, and serving pipelines.

3: Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

- ➔ Security is enforced at each boundary through encryption in transit/at rest and role-based access control before data moves to the next stage. Observability is enforced through logging, monitoring, and data quality checks at every transition, so failures or data drift are caught early rather than downstream.